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Programming of the Electronic Speed Controller in McMaster Autonomous Electric Vehicle

In this experiment, you will operate the motor drive system of the McMaster Autonomous Electric Vehicle (AEV). The motor drive system converts the electrical input power from the battery to mechanical output power to rotate the wheels and move the vehicle. Before you begin the experiments, please read the instructions in this manual carefully. **During the experiments, please follow all the instructions given in this manual.** Please note that the location, size, and models of some of the components shown in the figures in this manual might be slightly different from the components you are using in your project.

Motor Drive System Components

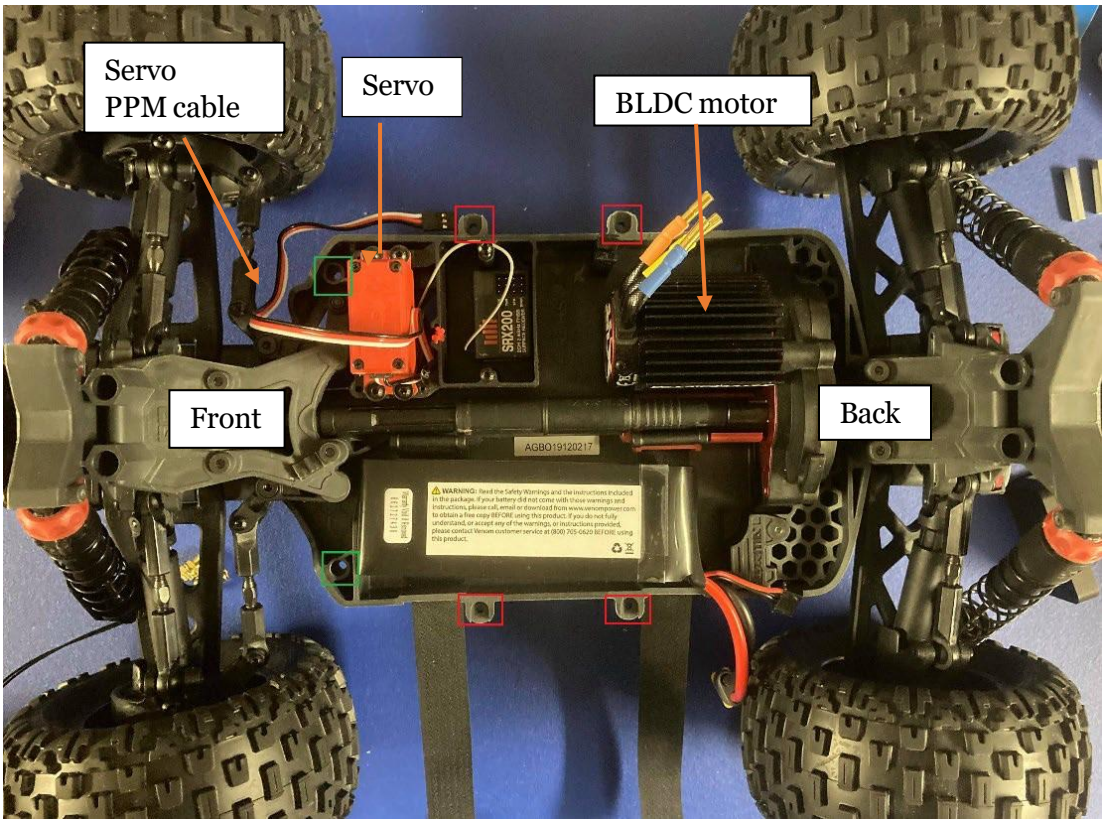

Figure 1: McMaster AEV chassis top view (Left: front of the car. Right: Back of the car)

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Figure 1 shows the McMaster AEV chassis top view. You can see the location of the battery, the Brushless DC (BLDC) motor, and the servo motor. The servo motor is used for steering the vehicle and the BLDC motor provides propulsion power. **In this lab, you will focus on the operation of the propulsion motor.**

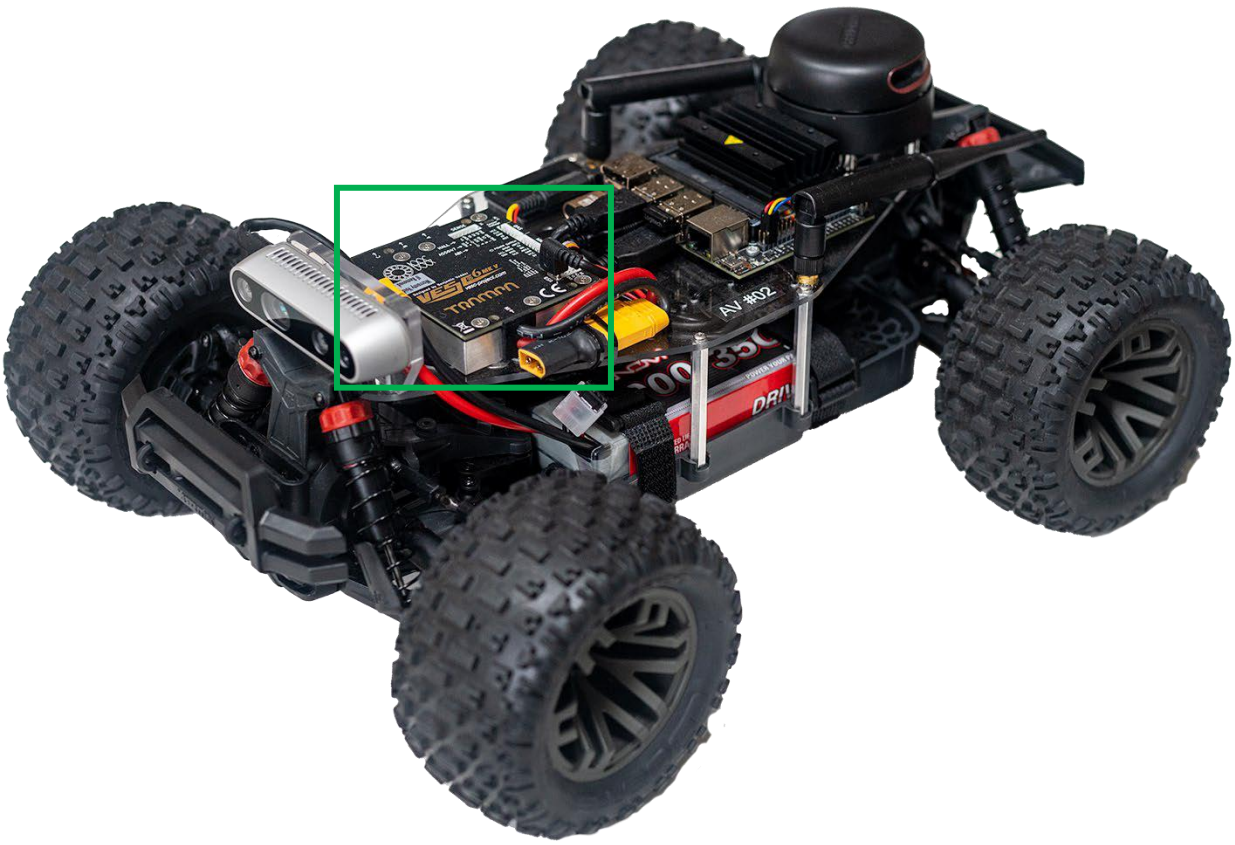

Figure 2: McMaster AEV and electronic speed controller

Figure 2 shows a complete version of a McMaster AEV. The BLDC motor is powered through an electronic speed controlled called VESC. Basically, VESC incorporates an inverter that drives the motor, sensing and control components and techniques to regulate the operation of the propulsion motor. VESC can be programmed using an open-source software called VESC Tool. VESC also incorporates motor control techniques such as field-oriented control (FOC) and BLDC control. Figure 3 shows a closer view of VESC hardware.

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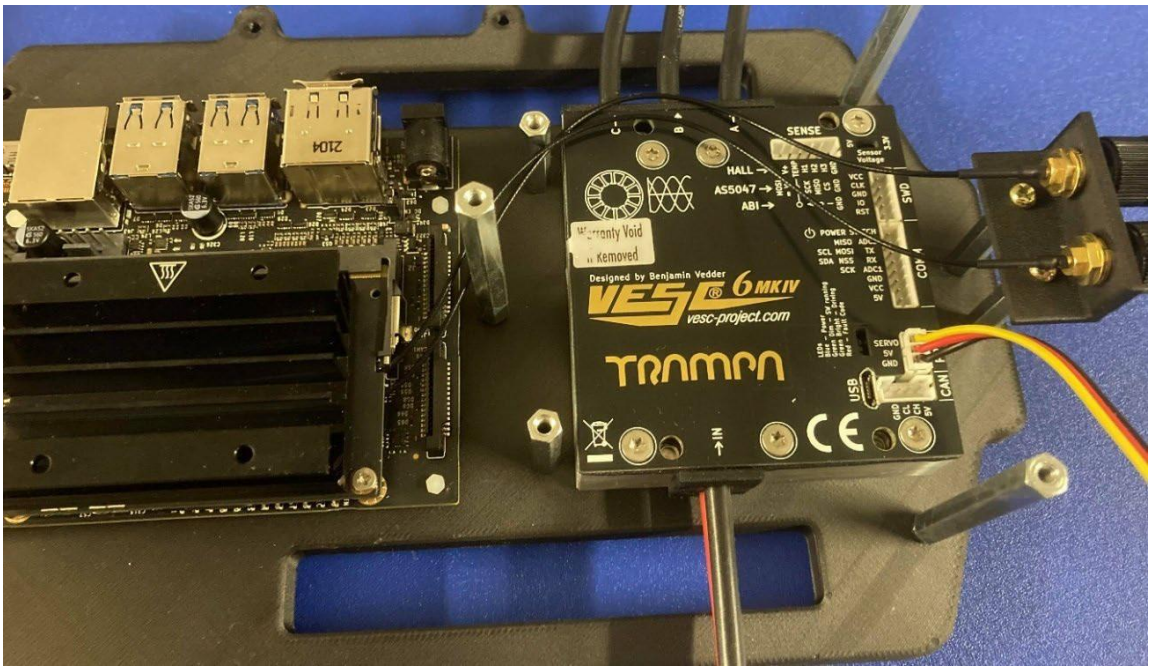

Figure 3: Electronic speed controller, VESC used in McMaster AEV

Important Safety Procedures to Follow

- Please be careful while you are in the lab and working with the vehicle.
- The experiments in this manual must be conducted in a laboratory environment under the supervision of teaching assistants (TAs). Please follow TA's instructions carefully.
- **Throughout the experiments given in this manual, the wheels of the McMaster AEV should be elevated or removed so that the vehicle does not move.**
- There must be no loose wires or metal pieces on your test bench.
- You must keep terminals and connections out of the way from accidental touching and from contact to test equipment or parts connected to other voltage levels.
- **You must not touch the vehicle or any components of the vehicle when the motor drive is powered, and the motor is operating.**
- Due to the heat generated in the motor drive system and as the motor consumes battery power, please do not take too much time for each measurement.
- No food or drinks are allowed in the lab.

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- **You must always check with the TA before you connect the battery to VESC and before you power up the motor drive.**
- **You must not adjust the motor drive parameters outside of the values provided in this manual.**
- If you encounter any issues with the motor drive operation, you must always check with the TA.
- Stop the operation of the motor first through VESC Tool, then disconnect the VESC Tool from VESC, and then disconnect the battery from VESC.

Useful Links on VESC and VESC Tool

Below, please find some links that can be helpful for you to get familiar with the operation and programming of VESC and setting up and use of VESC Tool. You **must review and study** the information in these links before and during the lab.

VESC Project home page: https://vesc-project.com/

VESC Hardware: https://trampaboards.com/1-x-vesc-6-mkv--p-27529.html

VESC Tool 2019 Tutorial: https://www.youtube.com/watch?v=Cs8eOPqRtgQ

VESC Tool App Tutorials: https://www.youtube.com/channel/UC6kQ4bL2xPoOio_oRgjdhQA/videos

Download VESC Tool: https://vesc-project.com/node/17

VESC Tool Documentation: https://vesc-project.com/documentation

VESC Tool Tutorial Videos: https://www.youtube.com/channel/UCcslH9JifCwInHV6mP5KGFQ

Objective 1: Review of the Lab Manual

Before starting your experiments, all members of your group must read and understand the lab manual. Then, you must explain to the TA what needs to be done in the lab, how you will conduct the experiments, and the safety precautions you need to consider. You cannot start the experiments before you answer your TA's questions.

It is mandatory that group members rotate after each objective. Each group member must contribute equally to the experiments. **In your report, you must state which group member lead the experiments in which objective.**

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Objective 2: Install and Setup VESC

Download and install the VESC Tool to your computer. **VESC Tool installation file is available on Avenue.** Follow the instructions that are available from the links provided above. Then, power up the VESC by connecting the VESC to the battery. In the McMaster AEV, VESC is connected to Jetson Nano via a USB cable. In order to connect VESC to your computer, unplug the USB cable for VESC from Jetson Nano and then use the adapter cable supplied in your kit to connect VESC to your computer. Make sure that the extension cable does not interfere with the wheels. **Please note that the instructions given in this manual might be slightly different from your vehicle depending on the operating system of your computer and the version of the VESC and VESC Tool.**

Open the VESC Tool and click AutoConnect as shown in Figure 4. If the VESC Tool is connected to the PC through serial port, **but in limited mode**, you might need to update the firmware.

Figure 4: Establishing connection with VESC Tool

In order to update the firmware, click Firmware at the left-hand side menu as shown in Figure 5. Your McMaster AEV might have an earlier or newer version of the VESC hardware. The New VESC Hardware version is MK6 and the Old VESC Hardware version is MK4. Both cases are shown in Figure 5 for reference. Pay attention to the hardware version you have in your McMaster AEV.

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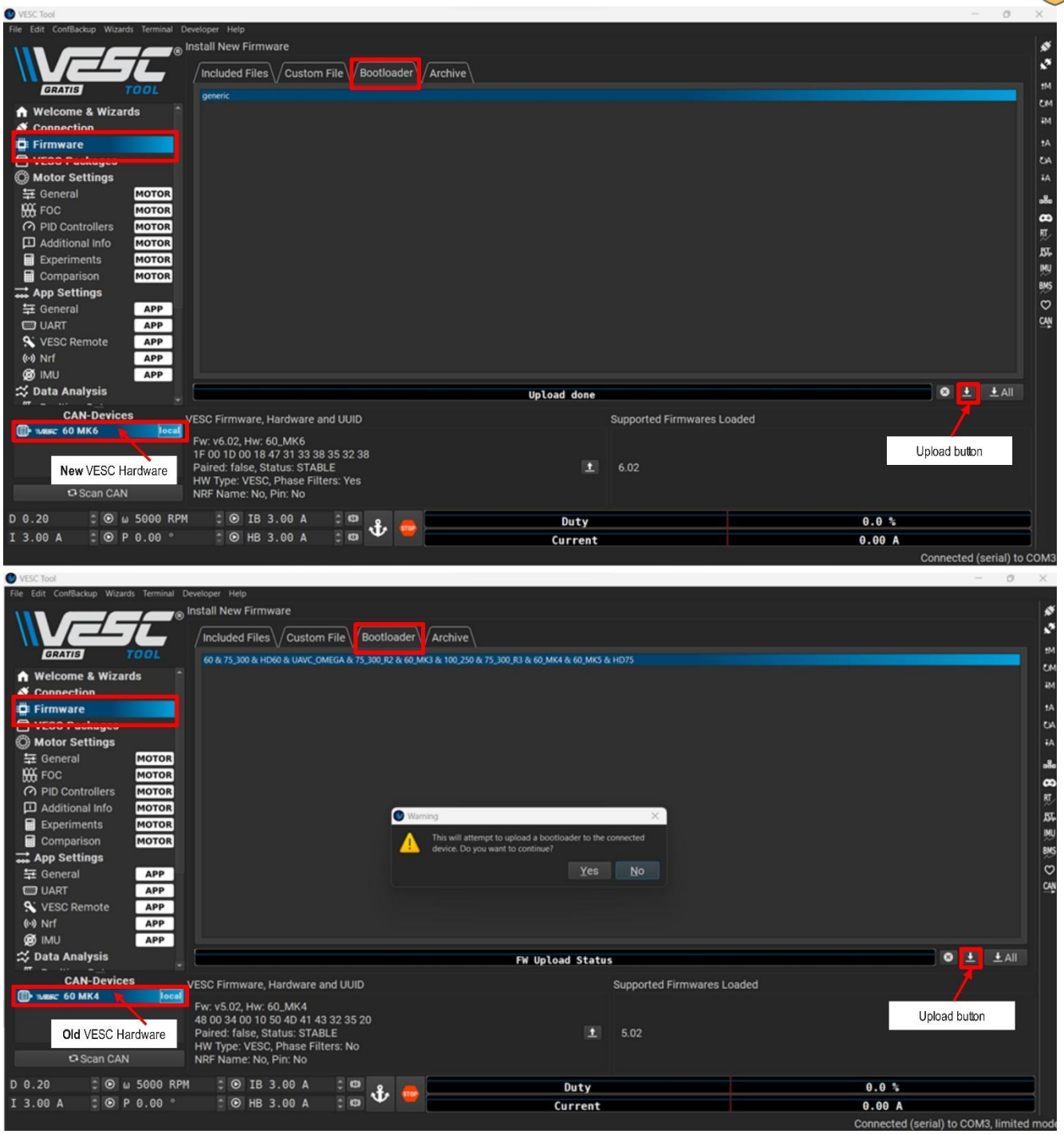

Figure 5: Setting up VESC firmware

Click Bootloader at the top of the Firmware screen. Select the bootloader option and click the upload button at the bottom of the screen. After that, a pop-up window will appear that asks you to confirm. Click Yes to continue. The progress bar labelled FW Upload Status will display the progress both as a green bar and as a percentage. When the upload process for the bootloader is complete, the bar will display Upload Done, and you can flash the new firmware.

As shown in Figure 6, click Included Files tab at the top of the screen and select the binary file that work with your VESC. It would probably be **"VESC_default_no_hw_limits.bin"** both for the Old and New VESC Hardware. Then, click the upload button again.

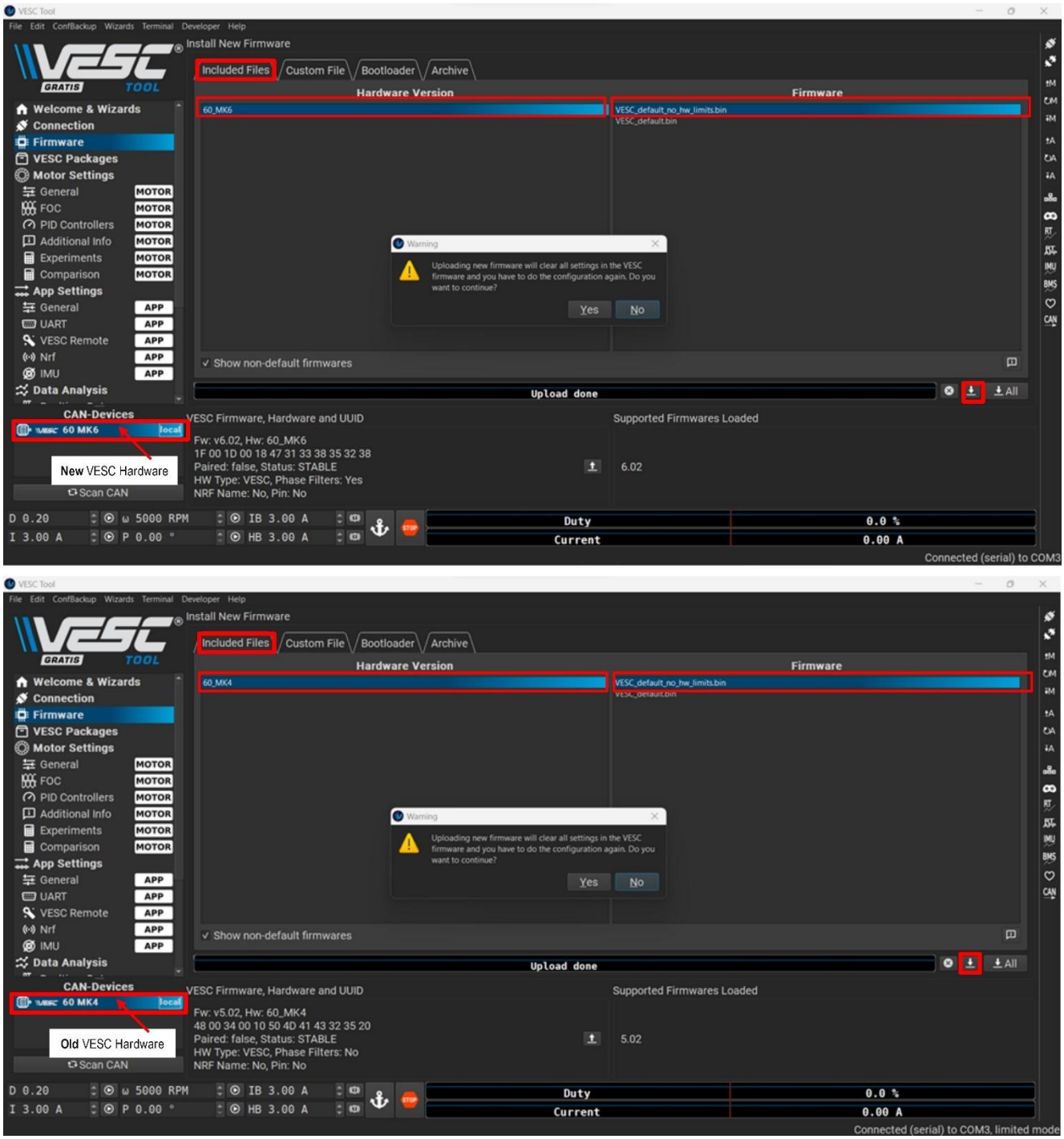

Figure 6: Uploading VESC firmware

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Click Yes again to start the flashing process. Again, a progress bar will appear. Wait for the flashing process to finish. When it is done, the Supported Firmware section will change to the correct version (e.g., 6.02) as shown in Figure 7. **A pop-up will appear warning that you should not power off VESC. As per the warning, after you have uploaded the firmware, wait at least 10 seconds before you disconnect and restart VESC.**

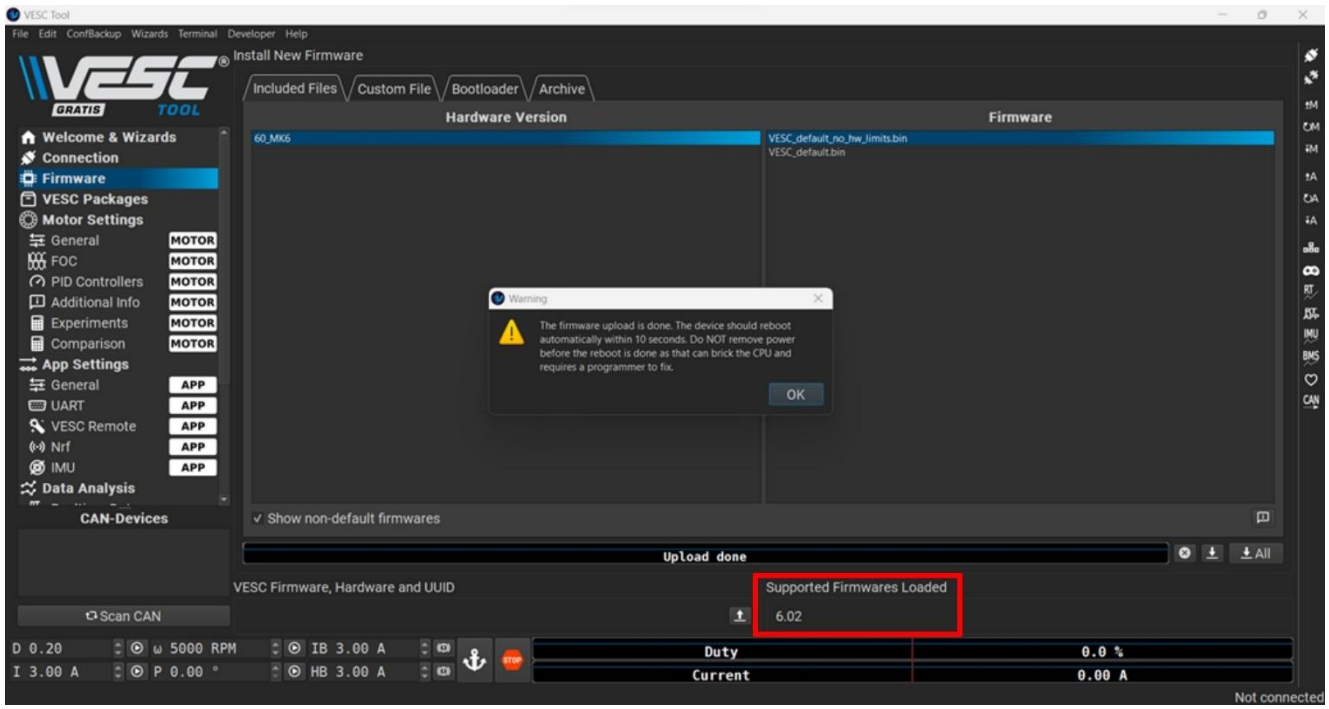

Figure 7: Firmware is updated

Once the firmware is updated, make sure that the VESC parameter **Enable Servo Output** is set to **True**. As shown in Figure 8, it can be found in **App Settings\General**. Once you set the parameter correctly, make sure to select **Write App Configuration** circled in red to write the parameter on VESC. Then, read it again using the read Current Setting circled in orange to ensure it is properly programmed.

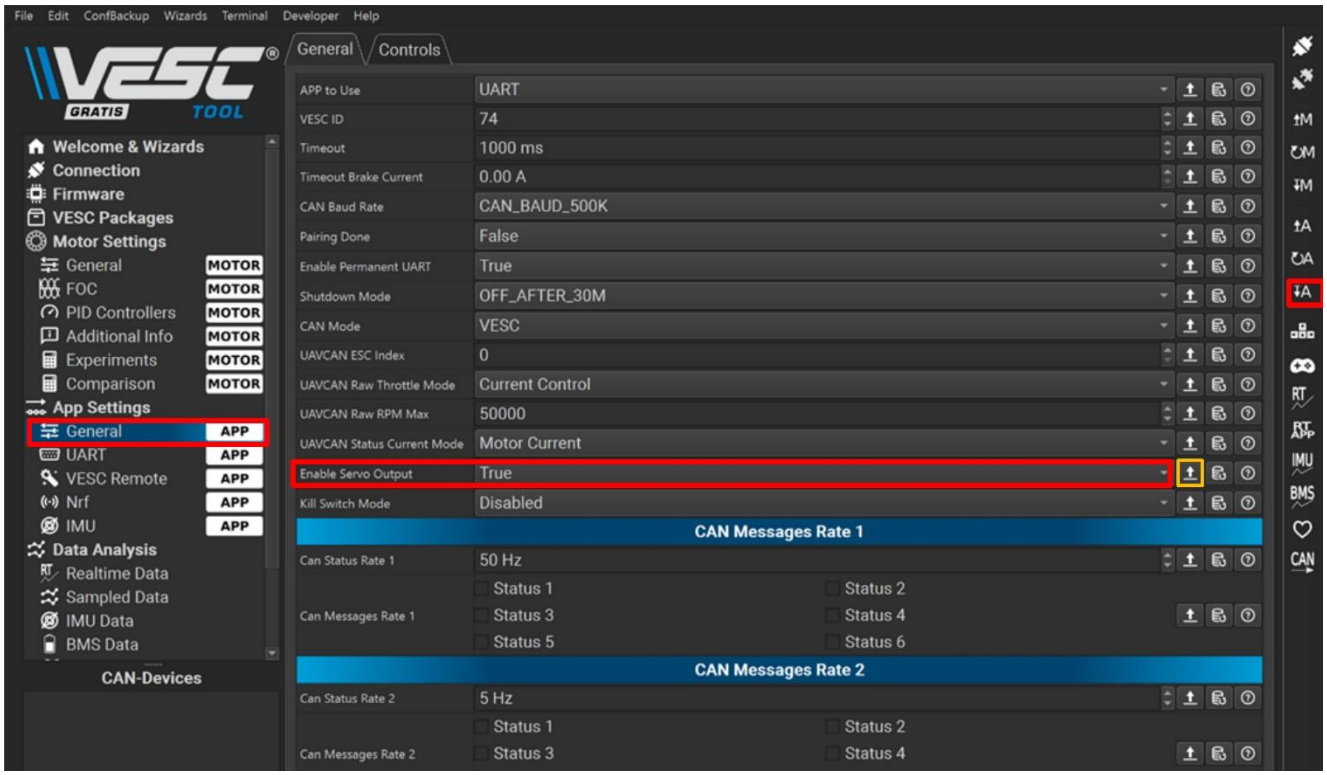

Figure 8: Enabling Servo Output parameter.

Before you can start motor parameter detection and calculation using the VESC Tool, you need to set the motor current limits properly as shown in Figure 9. <u>You should not set the motor current limits higher than what is given here.</u> Before you power up the motor drive, you must confirm the parameters with the TA. When completed, click the Write Motor Configuration button on the right-side menu circled in red. To confirm that you loaded the parameters correctly, click on the read current setting button circled in orange. If the parameters change, this means you have not written the motor configuration part correctly. If the numbers you applied stayed the same, this means you have the correct parameters and have written them to the motor correctly and can now proceed.

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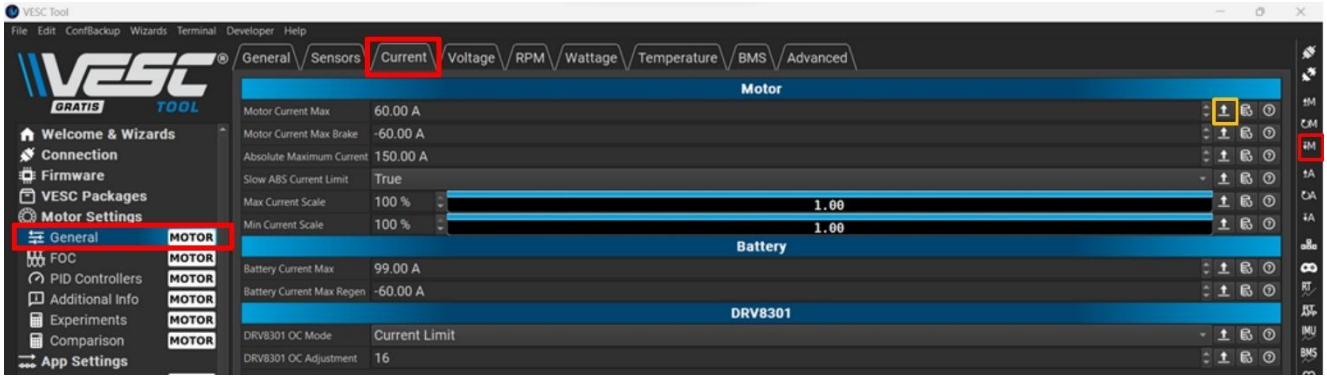

Figure 9: Setting up motor current limits

You also need to set the operating voltage limits of VESC based on the battery you are using. Figure 10 shows the voltage parameters set for a Lithium Polymer (LiPo) battery pack with three cells in series, with maximum 4.2 V per cell. In this example, the VESC tool calculated that VESC will enter the reduced power mode when the battery voltage is at 10.2 V and VESC will shut off at 9.3 V. This is essential to avoid over-discharging the battery. **Please check the battery parameters of your vehicle and check with the TA before entering the voltage parameters in VESC. After confirming the voltage parameters, click Apply button and then click the Load Motor Configuration button. Click on the read current setting button to confirm your parameters.**

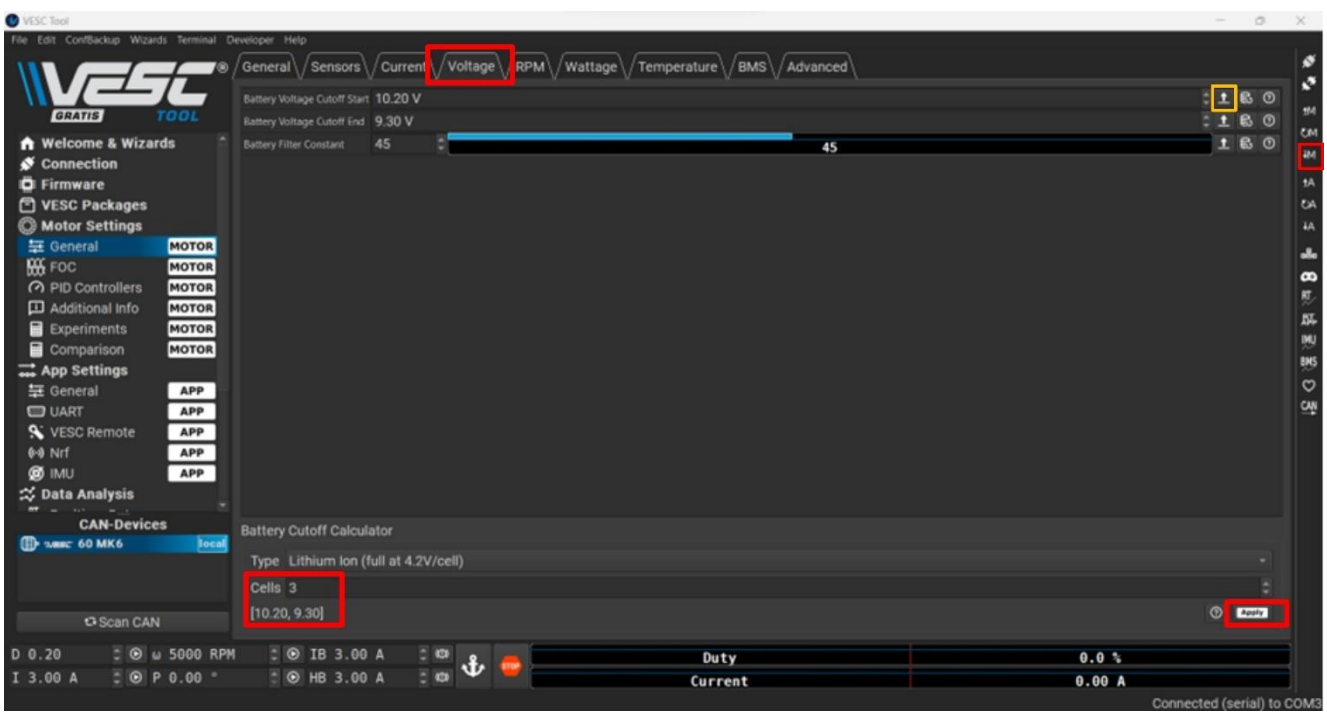

Figure 10: Setting up voltage parameters in VESC (shown for a 3S LiPo battery pack with maximum 4.2 V per cell; the parameters will be different for a different battery pack)

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Objective 3: Test and Report the Motor Parameters

You must first test the motor parameters using VESC. **Make sure that Field-Oriented Control (FOC) is selected as shown in Figure 11.** Please note that the motor will make some noise due to ramping of the current. **The wheels of the McMaster AEV should be elevated or removed so that the vehicle does not move. There should be no obstacle in contact with the wheels. Select RL, then select Lambda, and then select apply. Do not forget to click the Write Motor Configuration button and then confirm your parameters by clicking on the Read Current Setting button, so that you test and report the motor parameters.**

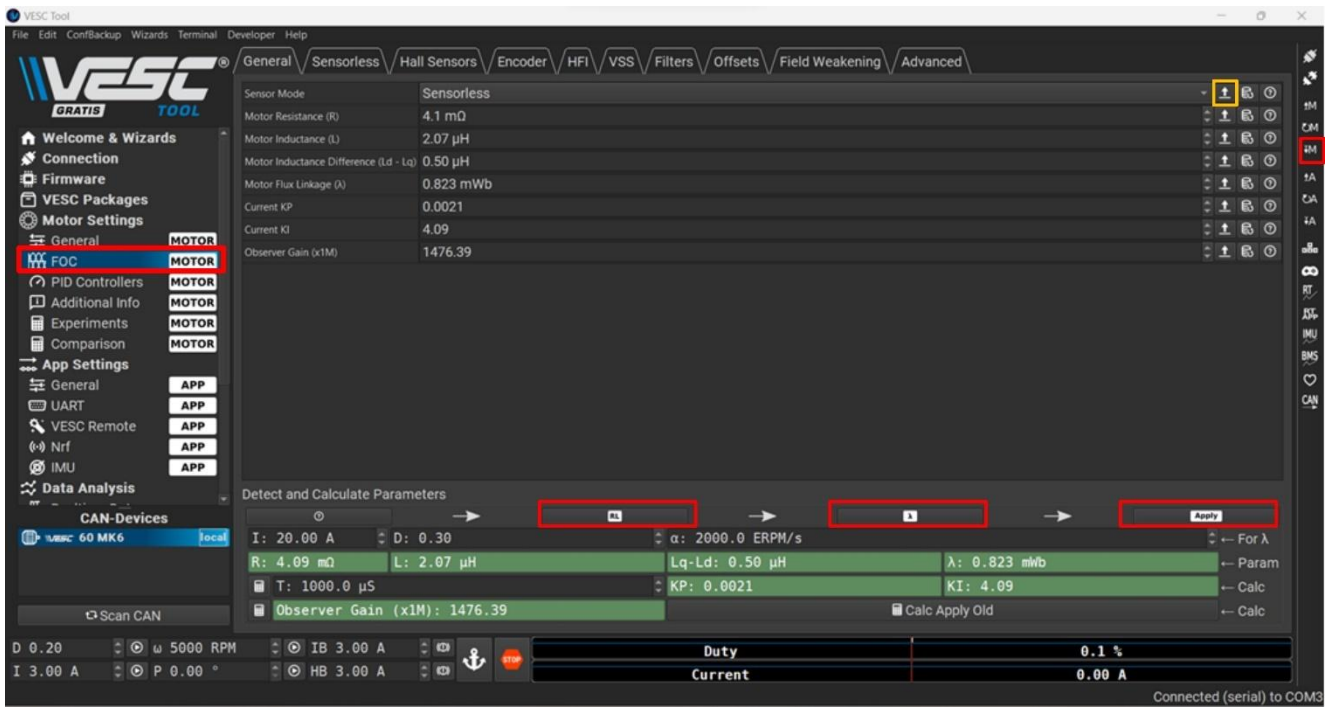

Figure 11: Testing motor parameters in VESC

VESC might not know the correct direction of rotation of the motor to move the vehicle forward and backward. If the default direction of rotation of the motor is the opposite of the forward direction of your vehicle, it needs to be fixed. As shown in Figure 12, select General menu from the side bar and select General tab. Click Refresh button and VESC Tool will identify the VESC connected to your computer. When you click Fwd button, VESC rotates the motor in the default forward direction. This is the direction of rotation when a positive speed or duty cycle command is applied. When you click Rev button, VESC rotates the motor in the default backward direction. This is the direction of rotation when negative speed command is applied.

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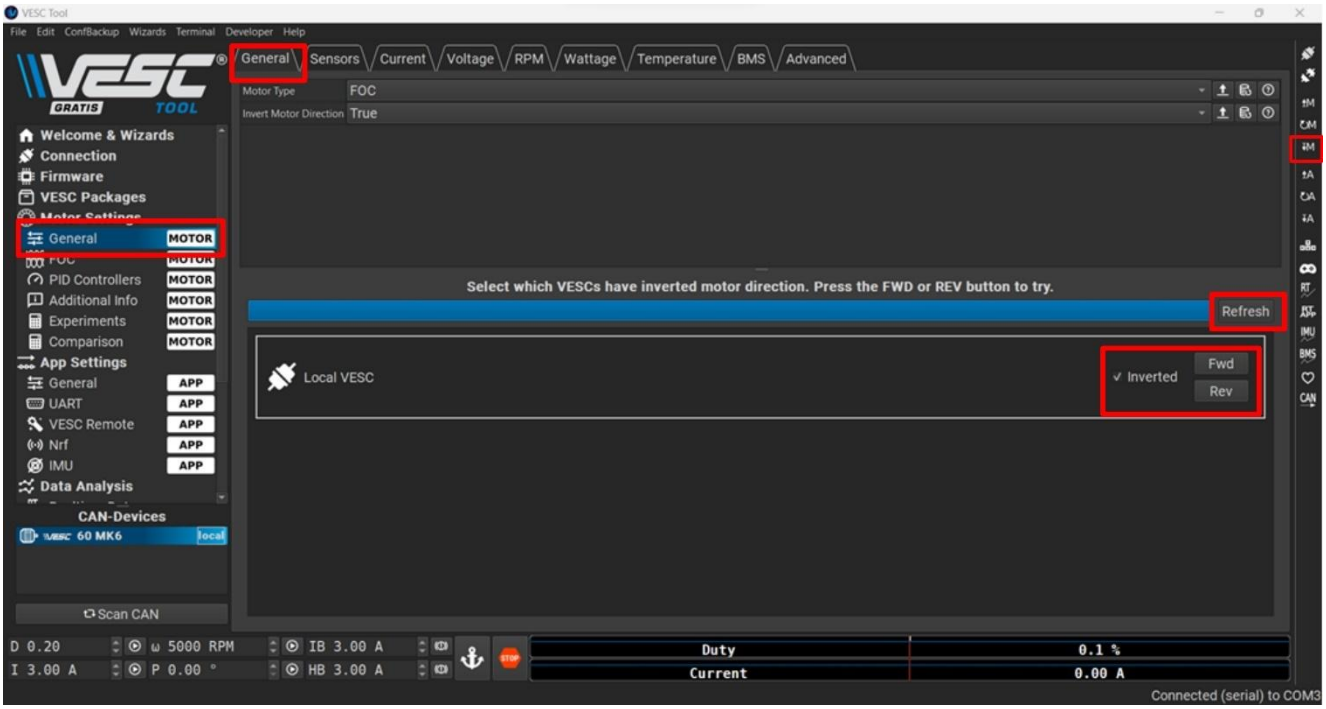

Figure 12: Changing the default direction of motor rotation

The front side of your car is where the camera is located. If the default forward rotation of the motor is the opposite of the forward direction of the car, check the Inverter dialog box and test the direction of rotation again. Make sure you click Write Motor Configuration button if you invert the direction.

Report the process and motor parameters calculated by VESC. Take a screenshot of the VESC tool when the identification process is completed.

Question 1: What do those parameters represent? Why are those parameters essential to control the motor?

Objective 4: Record Real-time Data and Set PID Parameters

To observe the real-time data, **make sure that the Stream Realtime Data button on the right-side menu is activated (turns into blue) as shown in Figure 13**. On the left-side menu, select Realtime Data under Data Analysis.

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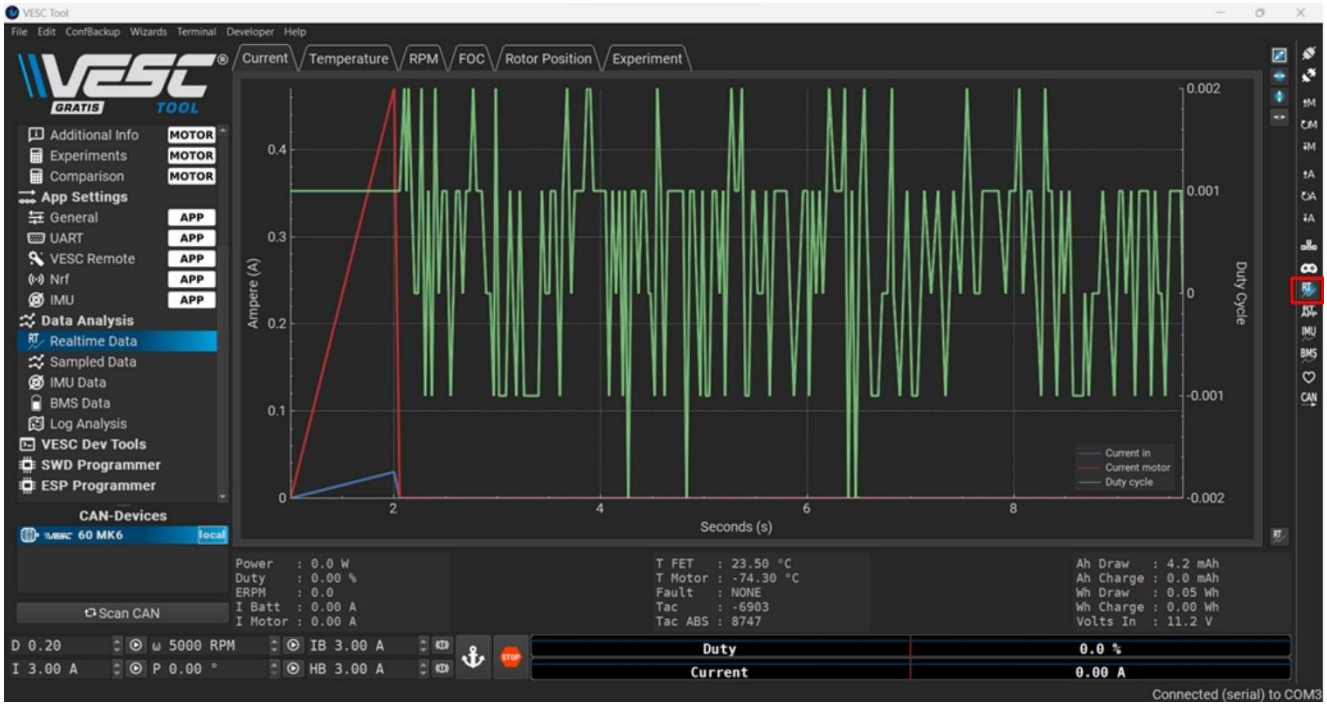

Figure 13: Enabling real-time data

Next, go to the PID Controller option under Motor Settings in the left hand-side menu as shown in Figure 14. Set the **PID speed controller parameters as K_p = 0.008, K_i = 0.00225, and K_d = 0.00035**. **Do not alter any other parameter.** Record the step response for speed. **Click the Load Motor Configuration button and then confirm your parameters by clicking on the read current setting button.**

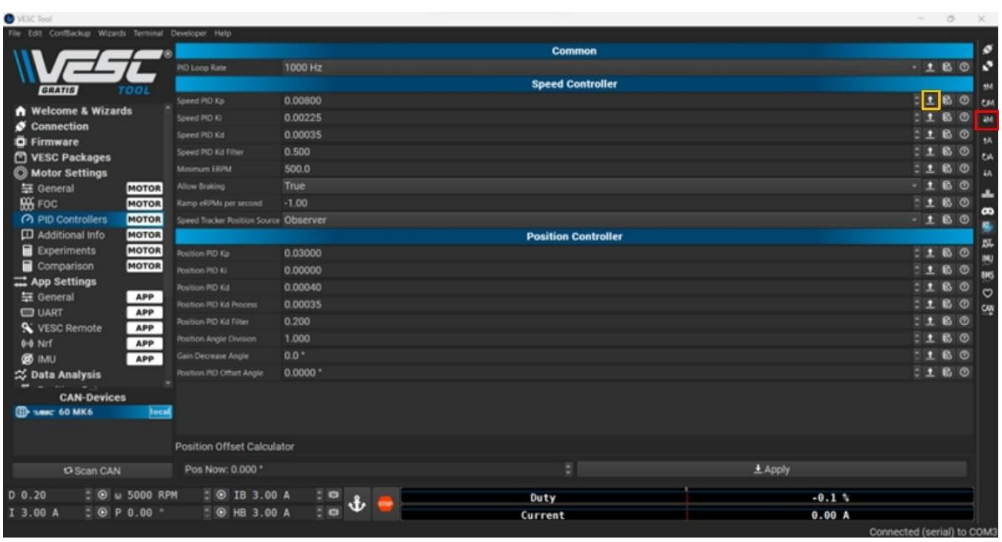

Figure 14: Setting up the PID parameters for speed controller.

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Question 2: Explain the importance of PID speed control parameters in relation to the operation of McMaster AEV?

The three types of input you can control are speed (ω), duty cycle (D), and current (I). In Figure 13, you can see the commands at the bottom of the screen.

The motor torque is related to the quadrature-axis current as discussed in the lectures. With a given current and, hence, motor torque, the motor speed will be related to the load torque. The load torque comes from the tractive power the vehicle needs. Please note that, in this experiment, the wheels are not touching the ground, or they are removed. Hence, the motor load torque is very low. Please be cautious with the commands that you apply to control and limit the motor torque. **Please follow the instructions below and confirm with the TA the parameters you would like to test before powering up the motor drive.**

The results you read from real-time data in VESC Tool can be logged and reviewed as well. On the menu bar, select Edit and then Preferences. A new window opens as shown in Figure 15. For the RT Data Log, Script Input Files, and Script Output Files, select a folder on your computer to save the generated .csv files that you will be creating.

Next, go to the Realtime Data option in the left-side menu, as shown in Figure 16. In order to log the data, click on the RT button on the right corner of the page labelled in green. A .csv file will be generated at the path you have chosen and will continuously log data. After you have finished operating the motor, simply uncheck the checkbox, and the .csv file will be saved. The name of the file will be a timestamp of the time you have enabled logging the data as shown in Figure 17. **Read through the appendix at the end of this lab manual** to plot the results from a .csv file.

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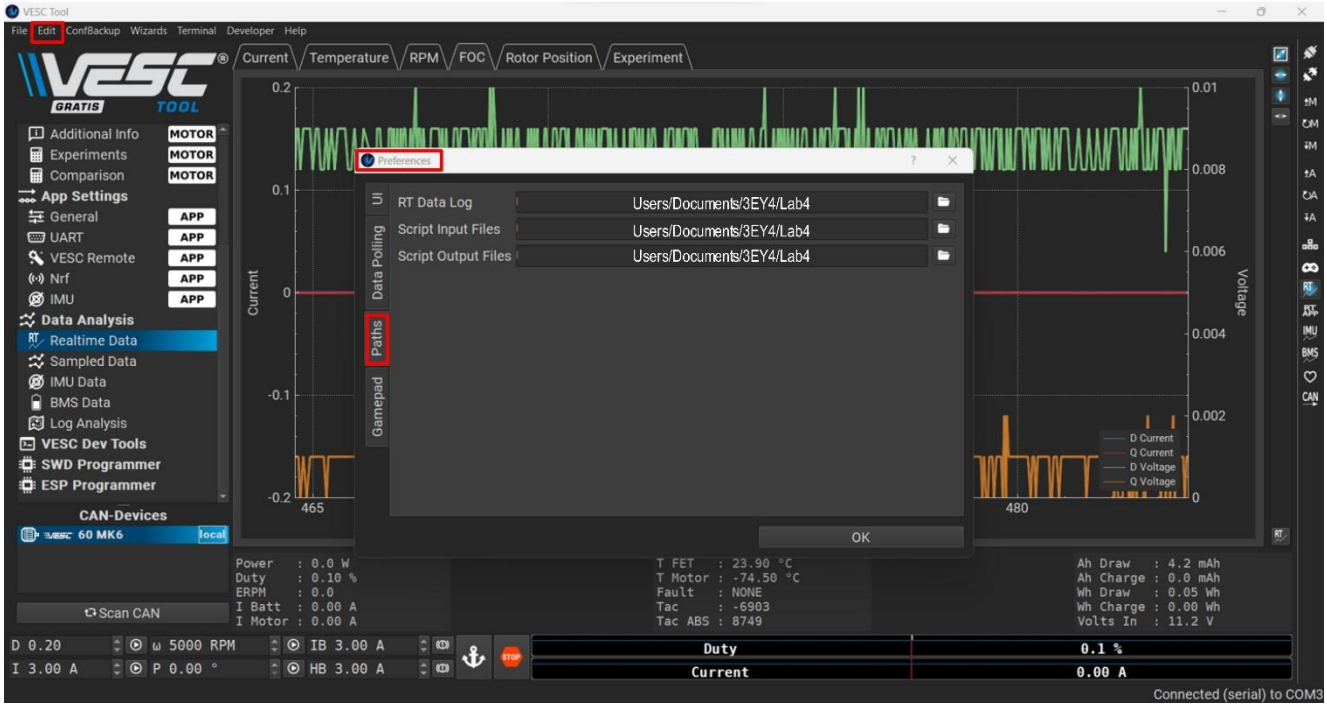

Figure 15: Creating a path to save the .csv files

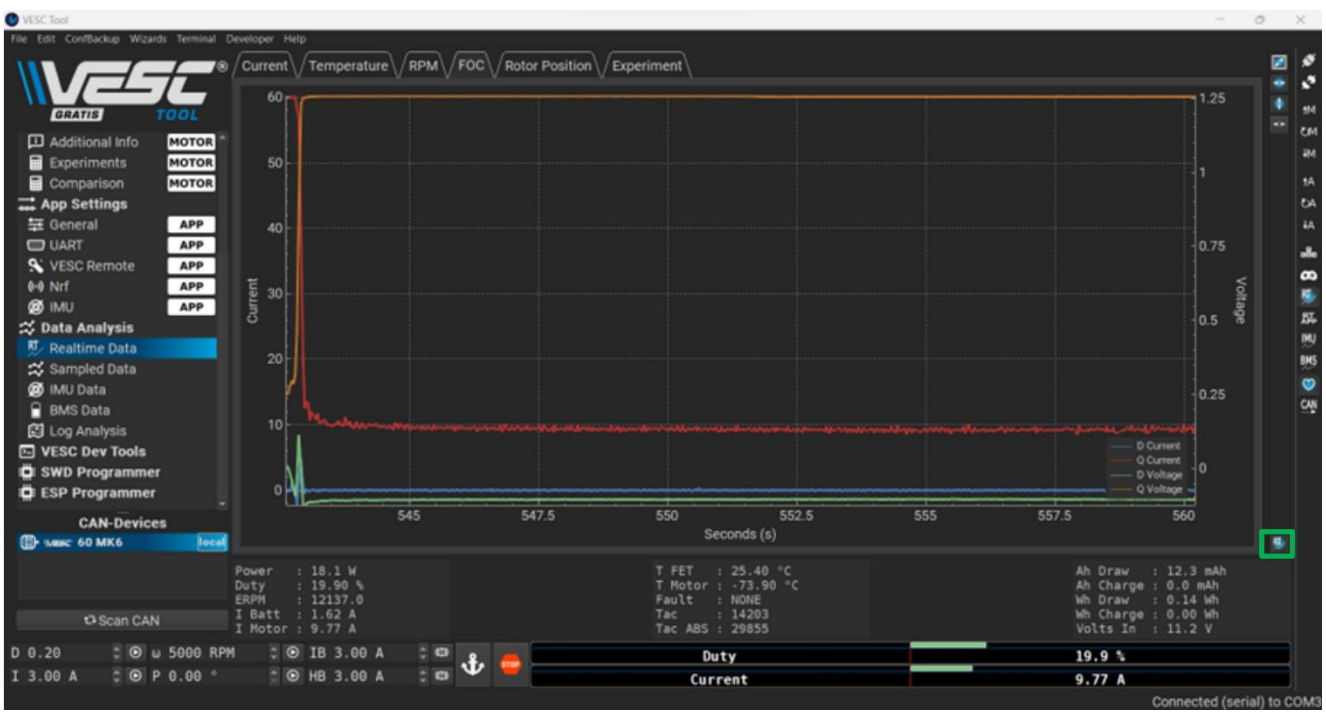

Figure 16: Data logging in VESC

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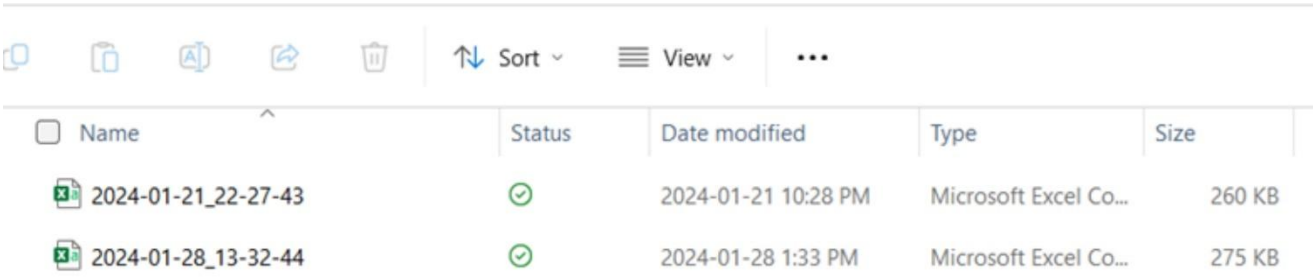

Figure 17: Timestamp of .csv file name

Question 3: First apply speed command to VESC and record how the speed, current and voltage change in time. Please change the speed gradually to a **maximum speed command of 7500 rpm**. (**Recommended:** start with a speed of 2500, then 5000 and finally 7500 rpm, Take a screenshot of Your RPM and FOC graphs. Log your data for each speed point and plot the graphs for RPM, FOC current (d- and q-axis currents), FOC voltage (d- and q-axis voltages).

Question 4: Next test the motor operation with duty cycle input. Please **maintain the duty cycle lower than 20%**. (**Recommended:** start with a duty cycle of 0.10, then 0.15 and finally 0.2, Take a screenshot of Your RPM and FOC graphs. Log your data for each speed point and plot the graphs for RPM, FOC current (d- and q-axis currents), FOC voltage (d- and q-axis voltages).

In this experiment, do not apply current command.

Question 5: What difference do you observe between the operations with the speed command and duty cycle command? Explain the difference using the real-time voltage, current, and speed waveforms.

Question 6: What is the definition of the d and q-axis currents? Why is the d-axis current always zero? Please check your lecture notes.